Chapter 1
Introduction

The theoretical description of strongly correlated electron systems remains one of the most challenging and fascinating problems in condensed matter physics. Unlike weakly interacting systems, where electrons behave as nearly independent quasiparticles, strong Coulomb interactions lead to the emergence of collective phenomena such as the Mott metal–insulator transition [1], non-Fermi-liquid behavior, unconventional magnetism [2] and high-temperature unconventional superconductivity [3]. Understanding these effects requires theoretical frameworks that go beyond single-particle approximations and simple mean-field descriptions. The treatment of strongly correlated electron systems is very challenging, even with modern computational approaches due to the exponential growth of the Hilbert space with system size and the presence of multiple competing energy scales.

Among modern approaches, dynamical mean-field theory (DMFT) [4] has established itself as a cornerstone for modeling local electronic correlations. By mapping a lattice problem onto a quantum impurity model [5], DMFT provides non-perturbative access to single-particle observables, including spectral functions and quasiparticle renormalizations. An important property of DMFT is that it becomes exact in the limit of infinite spatial dimensions, where non-local correlations vanish. In finite dimensions, however — and particularly in low-dimensional systems — spatial correlations can play a crucial role. This is especially relevant for unconventional superconductors, which are often effectively described as quasi-two-dimensional materials. It should be noted, though, that spatial correlations are not equally important for all collective phenomena: for example, the Mott metal-insulator transition is a collective effect that can still be accurately captured within DMFT. In contrast, d-wave superconductivity represents the opposite extreme, as its pairing symmetry fundamentally relies on the spatial structure of electronic correlations, which cannot be obtained from DMFT alone.

To overcome this limitation, so-called cluster extensions or diagrammatic extensions of DMFT have been developed (more on that in Ch. **??**), among which the dynamical vertex approximation (DΓA) [6, 7] is a state-of-the-art method. DΓA incorporates non-local correlations by taking two-particle irreducible quantities as building blocks and constructing ladder or parquet diagrammatic expansions to compute momentum- and frequency-resolved quantities. In this way, DΓA grants access to susceptibilities, self-energies and pairing instabilities beyond the local approximation, enabling the study of competing phases on equal footing [8].

While DΓA has been successfully applied to single-orbital systems [9, 10], many strongly correlated materials exhibit a pronounced multi-orbital character. The importance of this for a realistic description of materials is highlighted by the recent discovery of high-temperature superconductivity in compounds such as the bilayer nickelates $La_3Ni_2O_7$ [11], where orbital selectivity, Hund's coupling and intertwined instabilities strongly influence low-energy physics and superconducting pairing mechanisms [12]. Accurately describing these systems requires a multi-orbital generalization of the dynamical vertex approximation, which significantly increases computational and algorithmic complexity due to the higher dimensionality of the involved vertex functions. Such a formalism has been developed previously in e.g. Ref. [13]. In that work however, they did not consider superconductivity and the implications of multi-orbital correlations on it.

This thesis presents the implementation of a self-consistent multi-orbital ladder-DΓA framework applied to the multi-orbital Hubbard model, which is based on previous works, see Refs. [13–15]. The newly developed implementation is designed for numerical efficiency, scalability, and flexibility, enabling calculations for realistic multi-orbital systems. By exploiting vertex asymptotics, it allows us to access the correlated electronic structure at low temperatures which can provide us insights into the mechanisms relevant for superconductivity.

This work is structured as follows: Ch. **??** introduces a general theoretical background of condensed matter physics and how we model realistic materials. We mainly focus on the multi-orbital Hubbard Hamiltonian and its various simplifications, including the well-known Kanamori parametrization. We then introduce and develop the specific tools and concepts required for DΓA, including one- and two-particle Green's functions, susceptibilities, the Bethe-Salpeter equation and the Schwinger-Dyson equation in Ch. **??**. Furthermore, we also cover vertex asymptotics and explain their role in reducing numerical complexity while simultaneously significantly enhancing the capabilities of low-temperature calculations. Next, the ladder DΓA formalism and its connection to superconducting pairing instabilities via the Eliashberg equation are derived in Ch. **??**, and in Ch. **??** we describe the design and implementation of the code which was developed in the course of this thesis. We explain the parallelization strategy using MPI, detail the data structures and APIs and discuss optimization techniques such as symmetry exploitation and vertex asymptotics. This chapter also presents benchmark results, scaling analyses and a validation of the implementation against known reference cases. Lastly, we summarize the main contributions and outline possible future extensions in Ch. **??**.